ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Technical systems, gear trains, pulleys, work and power; parallel circuits, sensors and the NOT function; pneumatic symbols and a shuttle-valve circuit; motor control from two places and relay circuits; the design process, a fault-finding flowchart and maintenance.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. Draw all symbols, circuits and flowcharts neatly, using a ruler.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

QUESTION 1

1.1 Explain what a *technical system* is, and name its THREE main parts. (4)

1.2 FIGURE 1 below shows a simple gear train. Gear A is the driver.

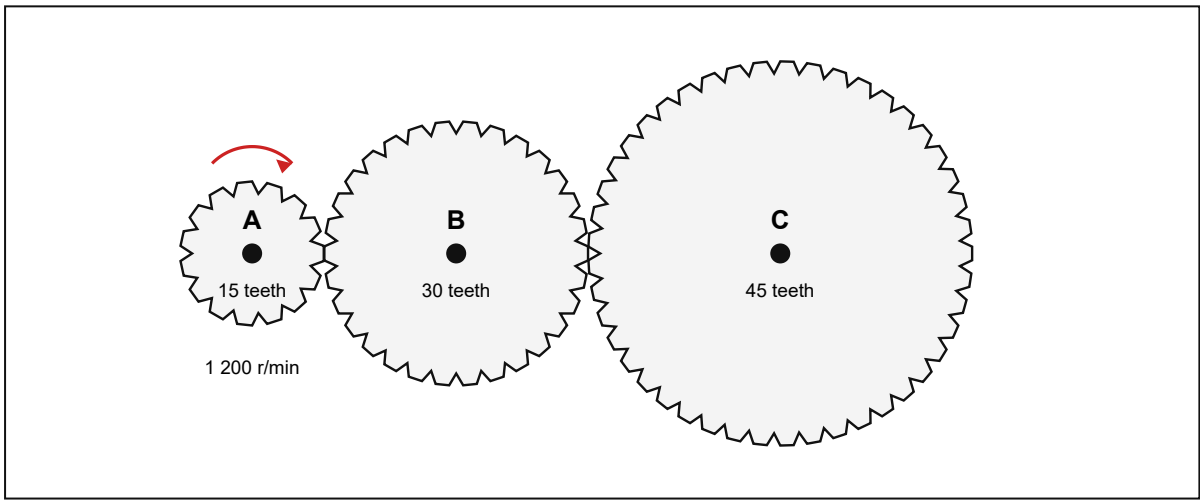


FIGURE 1

1.2.1 State the function of gear B. (2)

1.2.2 Calculate the speed of gear C. (3)

1.2.3 State the direction of rotation of gear C compared with gear A. (1)

1.3 Name TWO mechanisms that change rotary motion into linear (straight-line) motion. (2)

1.4 Explain the difference between a fixed pulley and a movable pulley. (2 × 2) (4)

1.5 A motor lifts a load of 200 N through a height of 3 m in 6 seconds. Calculate the following:

1.5.1 The work done (2)

1.5.2 The power developed (2)

[20]

QUESTION 2

2.1 Define the following terms:

2.1.1 Voltage (2)

2.1.2 Current (2)

2.2 Two resistors of 6 Ω and 3 Ω are connected in parallel across a 12 V supply. Calculate the following:

2.2.1 The total resistance (3)

2.2.2 The total current (2)

2.3 Name a suitable sensor for EACH of the following:

2.3.1 Detecting a metal object on a conveyor without touching it (1)

2.3.2 Measuring the temperature in an oven (1)

2.3.3 Detecting whether it is dark (1)

2.3.4 Detecting the end position of a pneumatic cylinder with a magnetic piston (1)

2.4 Draw a circuit in which a normally closed push button S1 and a lamp E1 form a NOT function. Also draw the truth table of the circuit. (5)

2.5 Write CPU in full and state its function in a controller. (2)

[20]

QUESTION 3

3.1 Draw the pneumatic symbol of EACH of the following:

3.1.1 Compressed-air supply (2)

3.1.2 Exhaust (without connection) (2)

3.1.3 Double-acting cylinder (2)

3.1.4 Shuttle valve (2)

3.2 FIGURE 2 below shows a pneumatic circuit in which a cylinder can be operated from two places.

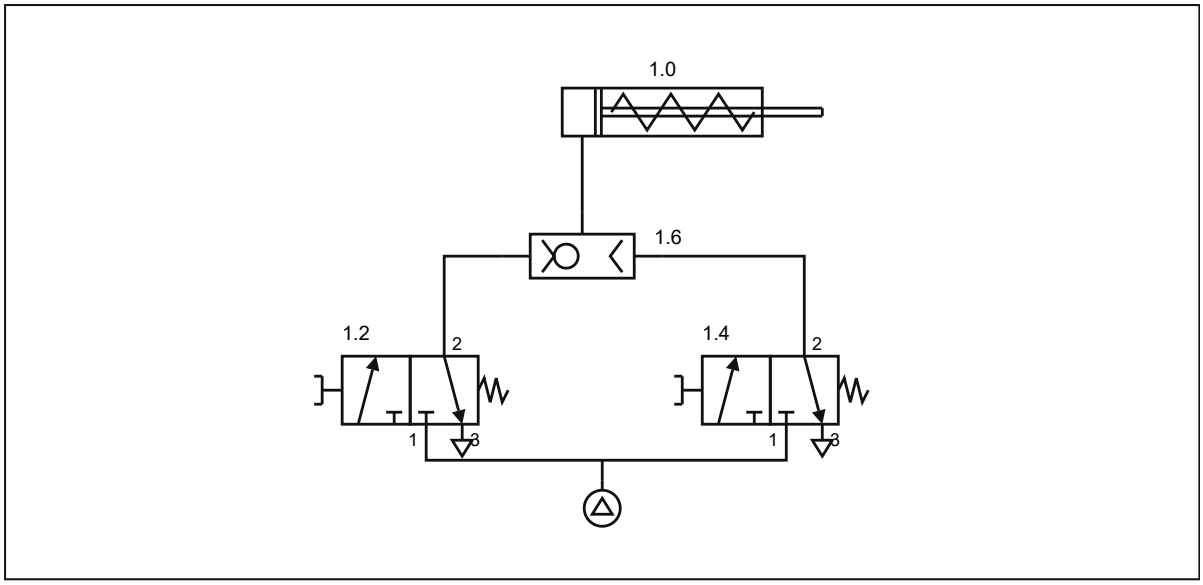


FIGURE 2

3.2.1 Name valve 1.6. (1)

3.2.2 Explain how the circuit works. (4)

3.2.3 Explain what would happen if valve 1.6 were replaced by a simple T-connector. (2)

3.3 Give THREE uses of pneumatics in industry. (3)

3.4 State the SI unit of pressure and ONE other unit that is often used for air pressure. (2)

[20]

QUESTION 4

4.1 Explain the function of EACH of the following:

4.1.1 A thermal overload relay (2)

4.1.2 An auxiliary contact of a contactor (2)

4.2 FIGURE 3 below shows a motor control circuit.

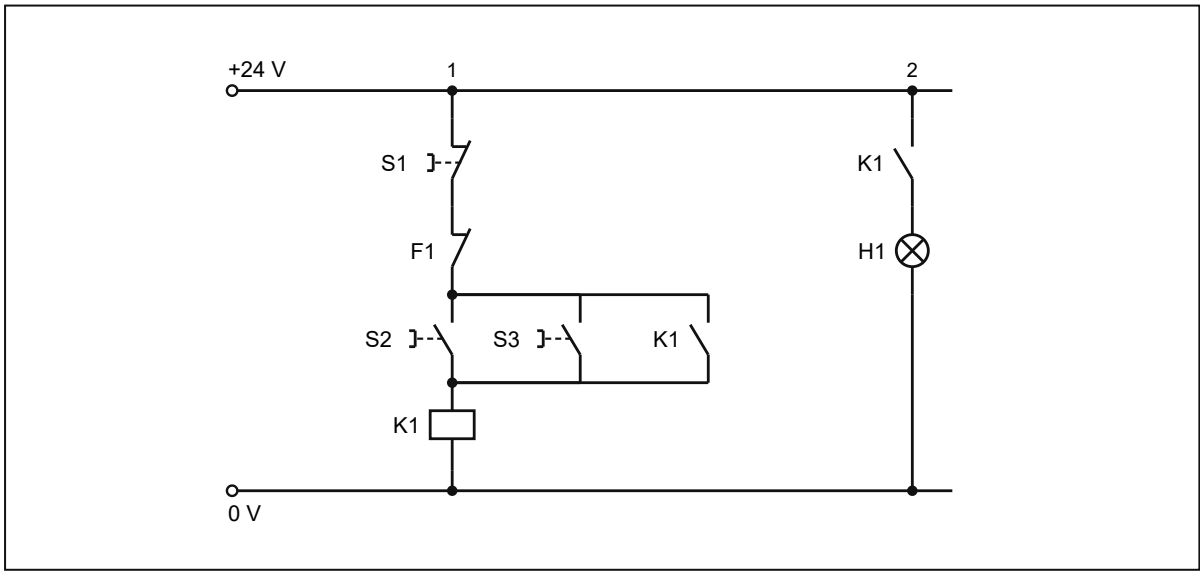


FIGURE 3

4.2.1 From how many places can the motor be started? (1)

4.2.2 How are S2 and S3 connected to each other? (1)

4.2.3 Explain what will happen if the holding contact K1 is faulty and does not close. (2)

4.2.4 State the function of F1. (1)

4.3 Draw a circuit in which pressing push button S1 energises relay K1, and a normally open contact of K1 switches on motor M1. (6)

4.4 Explain the purpose of electrical interlocking in a motor control circuit. (3)

4.5 Give TWO reasons why relays are used in control circuits. (2)

[20]

QUESTION 5

5.1 Explain why a prototype is built during the design process. (3)

5.2 List FOUR requirements that a good design must meet. (4)

5.3 A lamp in a control panel does not light up. Draw a fault-finding flowchart that checks the lamp, the fuse, the switch and the wiring. (6)

5.4 List FOUR checks that should be done during the daily inspection of a pneumatic machine. (4)

5.5 Explain why a maintenance schedule must be followed. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER